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Data Science FINAL PROJECT

REPORT

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Banking: Innovative Big Data
Management Techniques

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DECLARATION STATEMENT

This report is submitted in partial fulfilment of the requirement for the degree of Master of Science **in Data Science** at the University of Hertfordshire.

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Abstract

Banks lose billions each year to credit card defaults, yet many existing risk models still rely on approaches that struggle when the data is skewed heavily towards one outcome. This project tackled that problem directly by building and comparing three machine learning classifiers — Logistic Regression, Random Forest, and XGBoost — on a 30,000-client dataset from a Taiwanese bank. Because roughly four in every five clients in this dataset never defaulted, choosing the right evaluation metric was as important as choosing the right model. F1 Score was used as the primary metric throughout, since it penalises a model equally for missing actual defaulters and for raising false alarms.

The dataset presented a significant class imbalance of 78% non-default versus 22% default. To address this directly, SMOTE (Synthetic Minority Oversampling Technique) was applied to the training data, creating synthetic minority-class examples to balance the training distribution. Nine new behavioural features were engineered from the raw billing and payment columns to give the models richer predictive signals. Systematic hyperparameter tuning using GridSearchCV with 3-fold stratified cross-validation was applied to all three models optimising for F1 Score. ROC-AUC analysis was additionally performed to evaluate discriminatory power across all classification thresholds.

Results demonstrate that ensemble models outperform Logistic Regression for credit card default prediction. After applying SMOTE and hyperparameter tuning, XGBoost achieved the highest F1 Score (0.51), followed closely by Random Forest (0.50), while Logistic Regression achieved an F1 Score of 0.43. ROC-AUC analysis further confirmed the superiority of ensemble methods, with XGBoost achieving the highest AUC score of approximately 0.76. The most important predictors of default were repayment behaviour variables, particularly `avg_pay_delay` and `PAY_0`, while engineered behavioural features contributed substantially to model performance.

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1. Introduction

Credit card default imposes significant financial losses on banks, yet predicting which individual client will default in advance remains a genuinely difficult problem — one that existing scoring tools have not fully solved.

Traditional linear scoring tools cannot capture the complex, interacting patterns that precede default — a client with sustained high balances and frequent delays carries a fundamentally different risk profile from one who missed a single payment and recovered.

Machine learning addresses this by learning directly from historical patterns, with ensemble methods like Random Forest and XGBoost detecting subtle variable interactions that simpler models miss. This project builds a classification pipeline using a real dataset of 30,000 credit card clients from a Taiwanese bank. The task is to predict whether each client will default in the following month, based on six months of payment history, billing records, and basic demographic information. Three models are compared: Logistic Regression (as a linear baseline), Random Forest (an ensemble of decision trees), and XGBoost (a gradient boosting approach that has become widely used in financial prediction tasks).

One challenge this dataset presents is class imbalance — roughly 78% of clients never defaulted and only 22% did. This creates a trap: a model could score 78% accuracy just by predicting 'no default' for everyone, while completely failing at its actual purpose. To address this directly, SMOTE (Synthetic Minority Oversampling Technique) was applied to the training data, creating synthetic examples of the minority class to balance the training distribution. F1 Score is used as the primary evaluation metric throughout, since it balances the cost of missing real defaulters against the cost of incorrectly flagging safe clients. ROC-AUC analysis is additionally used to evaluate each model's discriminatory power across all decision thresholds, providing a threshold-independent performance measure.

2. Literature Review

The literature on machine learning for credit default prediction is substantial, but studies differ considerably in their handling of class imbalance, their choice of evaluation metrics, and the extent to which they combine feature engineering with systematic model comparison. This review critically examines the most relevant prior work and situates the present project within that landscape.

2.1 The Dataset and Its Origins

Yeh and Lien (2009) assembled the dataset used in this project, originally to test whether data mining approaches could outperform the scoring tools that Taiwanese banks were using at the time. Their study compared six approaches including discriminant analysis, logistic regression, and neural networks, finding that the neural network approach produced the most accurate probability of default estimates. However, a critical limitation of their evaluation was that they optimised for overall accuracy, which, as this project demonstrates, is deeply misleading when the classes are as imbalanced as they are here. Their reported accuracy figures of around 81-82% are nearly indistinguishable from what a trivial majority-class classifier would achieve, and their paper does not report recall or F1 Score for the minority class.

2.2 Why Class Imbalance Handling Matters

Brown and Mues (2012) conducted a systematic comparison of classifiers on credit datasets with heavy class imbalance and found that standard models consistently favour the majority class at the expense of correctly identifying the minority — which in this context means defaulters. Their central recommendation was for evaluation frameworks that give equal weight to performance on both sides of the outcome, and they specifically highlighted F1 Score and AUC as the appropriate alternatives to accuracy. Importantly, they also found that imbalance-handling techniques — when properly applied only to training data — could substantially improve minority-class recall without compromising the integrity of evaluation.

This project addresses this by applying SMOTE exclusively to the training set, preventing data leakage into evaluation.

He and Garcia (2009) surveyed imbalanced learning techniques and found that SMOTE typically outperforms undersampling and matches cost-sensitive approaches by generating informative synthetic examples that improve the minority-class decision boundary, directly motivating its use here.

2.3 Ensemble Methods and Gradient Boosting

Lessmann et al. (2015) compared more than forty classification algorithms across credit scoring benchmarks and found that ensemble methods — particularly gradient boosting and bagging approaches — consistently outperformed linear models and single-tree methods. Their study is methodologically important because it also demonstrated that the performance advantage of ensemble methods widened when class imbalance was properly handled, and when evaluation was conducted on genuinely held-out test sets rather than cross-validation folds applied to the full dataset. The finding that accuracy is insufficient as an evaluation criterion is perhaps their most practically significant contribution.

Chen and Guestrin (2016) introduced XGBoost as a gradient boosting system built for speed and scalability on tabular data. Its key advantages over earlier gradient boosting implementations include built-in L1 and L2 regularisation, handling of sparse inputs, and an efficient tree-splitting algorithm. Its regularisation properties make it less prone to overfitting than unregularised ensembles — a key advantage in credit risk settings.

Breiman (2001) introduced Random Forests; their Gini-based feature importance scores provide interpretable variable rankings used in this project's analysis.

2.4 The Value of Engineered Features

Xia et al. (2018) demonstrated that features constructed from raw billing and payment data — rather than just the original columns — can make a meaningful and statistically significant difference to predictive accuracy in credit scoring models. Their study applied a boosted decision tree approach with Bayesian hyperparameter optimisation and found that aggregate features capturing repayment consistency and credit utilisation over multiple months were consistently among the most important predictors. Critically, they also found that the combination of feature engineering and ensemble methods outperformed either approach in isolation — a finding directly relevant to the present project, which applies both simultaneously.

The nine engineered features in this project — including average payment delay, pay-to-bill ratio, and credit utilisation ratio — capture aggregate behavioural patterns the raw monthly columns cannot directly express, motivated by Xia et al. (2018).

2.5 Comparison with the Present Work

Taken together, the reviewed literature motivates the specific methodological choices made in this project: F1 Score and ROC-AUC as primary metrics (in response to Brown and Mues, 2012), SMOTE applied exclusively to training data (He and Garcia, 2009), ensemble methods as the preferred modelling approach (Lessmann et al., 2015), and engineered behavioural aggregates alongside raw features (Xia et al., 2018). The pipeline presented here applies all four components simultaneously in a single reproducible evaluation on the Yeh and Lien (2009) benchmark dataset, enabling direct and transparent comparison with prior work.

3. Research Questions and Objectives

3.1 Research Question

This project is guided by the following primary research question:

Which machine learning algorithm performs best among Logistic Regression, Random Forest, and XGBoost for predicting credit card default, in terms of Accuracy, Precision, Recall, and F1 Score?

3.2 Objectives

To address the research question, the following specific objectives were defined:

- To perform exploratory data analysis on the credit card client dataset to understand the distribution, relationships, and class imbalance of key variables.
- To preprocess the dataset, including removal of the identifier column, correction of irregular categorical values, and standardisation of numeric features for linear models.
- To engineer nine new predictive features derived from billing and payment history to enrich the feature space and capture aggregate behavioural patterns.
- To apply SMOTE to the training data to address the 78/22 class imbalance, and to evaluate its impact on model recall separately from the effect of hyperparameter tuning.
- To train and evaluate three machine learning models — Logistic Regression, Random Forest, and XGBoost — across baseline (no SMOTE), SMOTE baseline, and SMOTE-tuned conditions.
- To apply GridSearchCV-based hyperparameter tuning with 3-fold stratified cross-validation, optimising all models for F1 Score.
- To compare model performance using Accuracy, Precision, Recall, F1 Score, confusion matrices, and ROC-AUC, and to draw conclusions about the relative contribution of each pipeline component.

4. Ethical Considerations

The dataset is publicly available through the UCI Machine Learning Repository and contains no personal identifiers — all records are anonymised numeric codes. This project was conducted for academic purposes only, with all sources duly cited and all work the author's own.

One consideration worth noting is that the dataset includes gender, education, and marital status. In a real-world credit decision system, using demographic variables as model inputs raises legitimate fairness concerns and would require disparate impact analysis and regulatory review prior to deployment. For this academic benchmark, these features are retained as part of the original published dataset, with the understanding that their inclusion reflects the benchmark's composition rather than an endorsement of their use in production systems.

5. Dataset Description

5.1 Source

The dataset comes from a Taiwanese bank and was assembled by Yeh and Lien (2009) as a benchmark for credit risk research. It is publicly available through the UCI Machine Learning Repository. It covers six consecutive months — April through September 2005 — and contains 30,000 client records stored in an Excel spreadsheet format with headers on row 2.

5.2 Features

The dataset contains 25 columns, of which 24 are features and one is the target variable.

Feature	Data Type	Description
ID	Integer	Unique client identifier (removed in preprocessing)
LIMIT_BAL	Integer	Credit limit in New Taiwan Dollars (NT\$)
SEX	Integer	Gender: 1 = Male, 2 = Female
EDUCATION	Integer	Education level: 1=Graduate, 2=University, 3=High School, 4=Others
MARRIAGE	Integer	Marital status: 1=Married, 2=Single, 3=Others
AGE	Integer	Client age in years (range: 21–79)
PAY_0 – PAY_6	Integer	Repayment status for the past 6 months (-2 to 8)
BILL_AMT1–6	Integer	Bill statement amount for the past 6 months (NT\$)
PAY_AMT1–6	Integer	Previous payment amount for the past 6 months (NT\$)
default payment next month	Binary	Target variable: 1 = Default, 0 = No Default

Table 1: Dataset Features and Descriptions

5.3 Target Variable

The target variable is whether a client defaulted on their credit card payment the month after the observation window closed. Just under a quarter of clients in the dataset defaulted (6,636 out of 30,000, or approximately 22%), while the remaining 23,364 did not. This 78/22 imbalance is large enough to cause serious problems for any model evaluated solely on accuracy.

5.4 Key Statistical Properties

Credit limits range from NT\$10,000 to NT\$1,000,000, with a mean around NT\$167,000. The average client is in their mid-thirties. The payment status variables (PAY_0 through PAY_6) mostly sit at or near zero — meaning most clients were paying on time — but there is a tail of clients with delays stretching up to eight months. Bill amounts show considerable spread, and a small number of clients carry negative balances indicating they had overpaid. No missing values were found anywhere in the dataset.

Correlation analysis identified PAY_0 (most recent payment status) as the feature most strongly correlated with the target variable ($r = 0.32$). The bill amount variables (BILL_AMT1 through BILL_AMT6) showed high inter-correlation with each other ($r > 0.90$ in many pairwise cases), indicating substantial information redundancy across months. This redundancy motivates the aggregation approach taken in feature engineering.

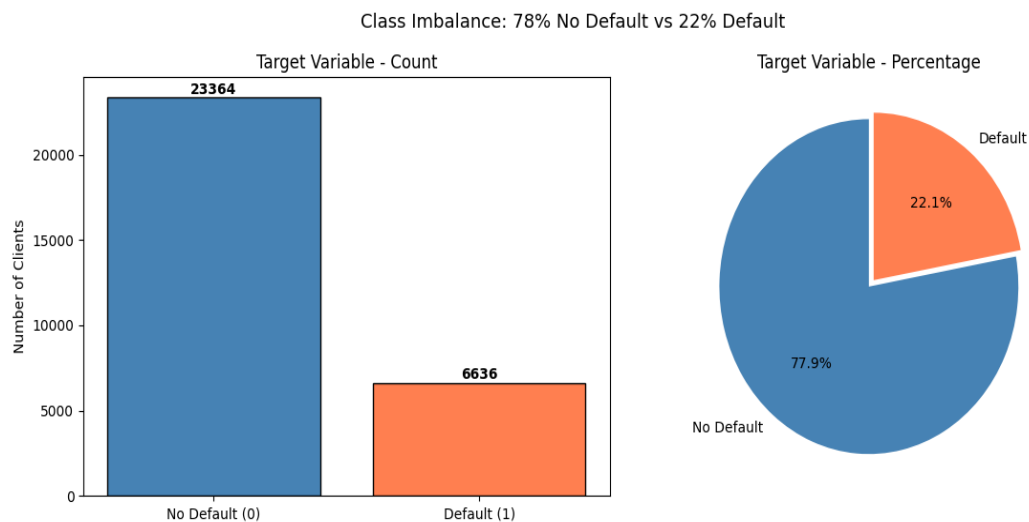
6. Methodology

The methodology followed a structured machine learning pipeline: EDA and preprocessing, feature engineering, train-test splitting, SMOTE application, model training, hyperparameter tuning, and systematic evaluation. Each step is described in detail below.

6.1 Exploring the Data

Exploratory Data Analysis was performed to understand the structure, distribution, and relationships within the dataset:

- Missing value analysis: The dataset was checked using `pandas isnull().sum()`. No missing values were found across any column, confirming completeness. No imputation was required.
- Target distribution analysis: The class distribution was examined using `value_counts()` and visualised with a bar chart and pie chart. The analysis confirmed a significant imbalance of approximately 78% non-default to 22% default, with 23,364 non-defaulters and 6,636 defaulters in the full dataset.



Figure= 1: Target Variable Distribution

- Distribution analysis of key features: Histograms were generated for `LIMIT_BAL`, `AGE`, `BILL_AMT1`, and `PAY_AMT1`. Credit limit distribution is right-skewed, with the majority of clients holding limits between NT\$50,000 and NT\$300,000. Age distribution is approximately normal, centred in the mid-thirties. Bill amounts are strongly right-skewed, reflecting that a small number of clients carry disproportionately large balances.

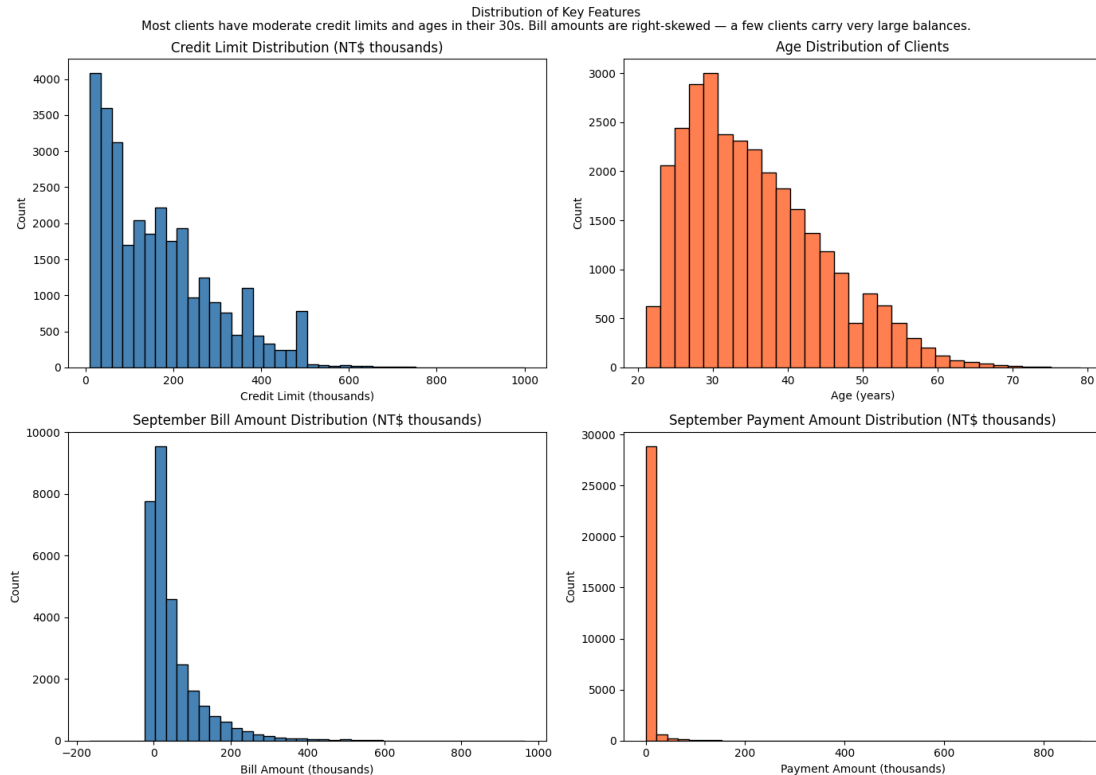


Figure 2: Distribution of Key Numeric Features

- Default rate by PAY_0: A bar chart revealed a clear non-linear threshold — clients current on payments had default rates of 10–15%, while those with delays of two or more months approached 50–70%, explaining why linear models underperform on this task.

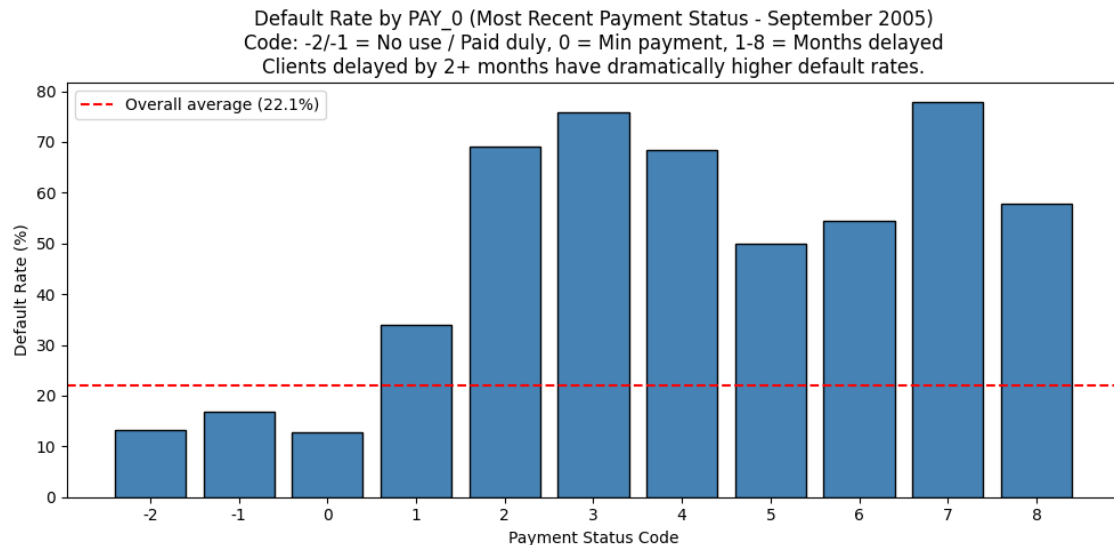


Figure 3: Default Rate by Payment Status (PAY_0)

- Correlation heatmap: A Pearson correlation matrix was constructed for all numeric features. PAY_0 showed the highest correlation with the target ($r = 0.32$), followed by PAY_2 and PAY_3. The BILL_AMT columns showed very high mutual correlations ($r >$

0.90), indicating information redundancy that the feature engineering step is designed to reduce through aggregate

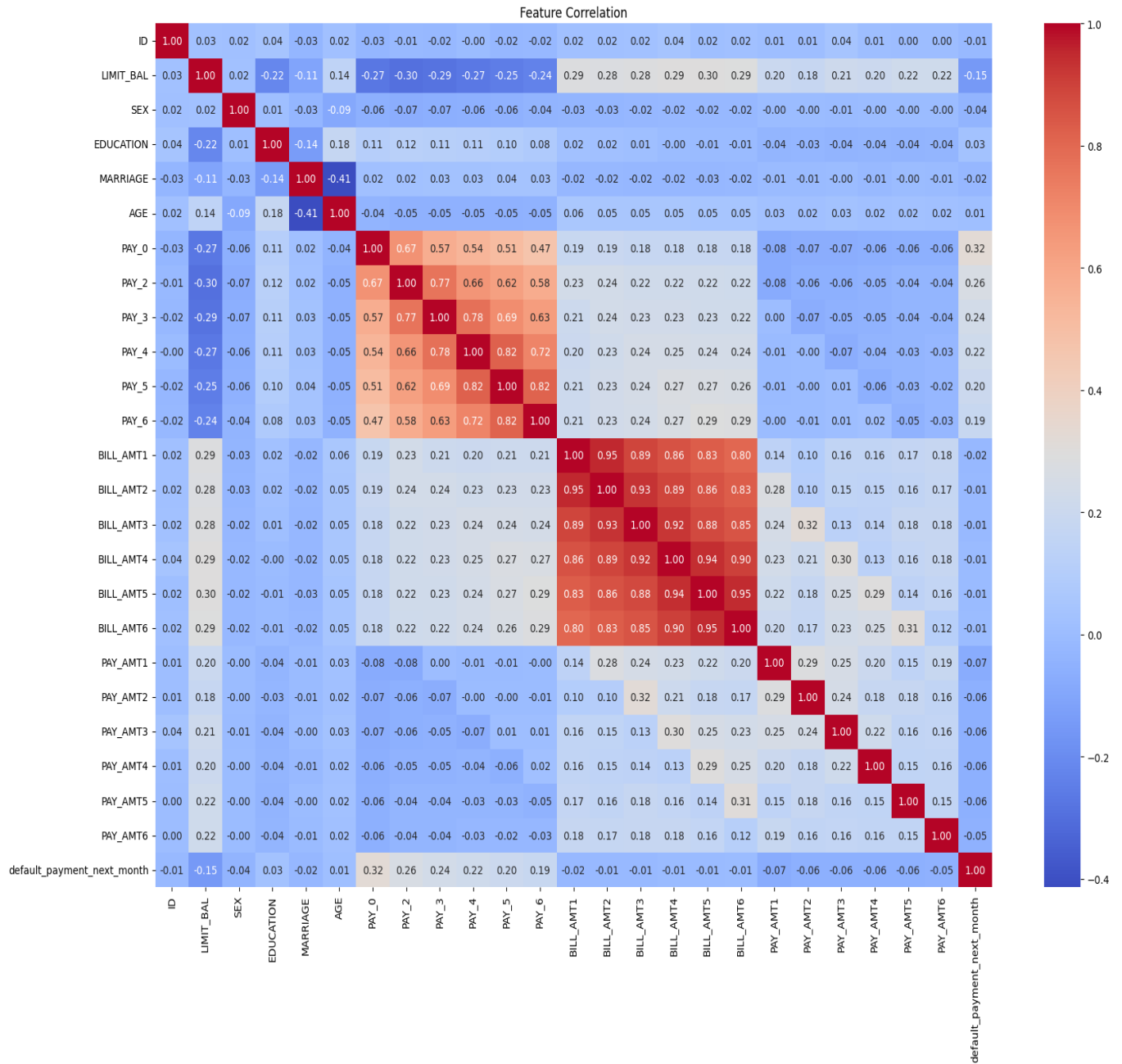


Figure 4: Correlation Heatmap

The correlation analysis directly informed feature engineering decisions: the high inter-correlation of BILL_AMT column motivated aggregation into summary statistics rather than retaining all six individually.

6.2 Data Cleaning and Preprocessing

The ID column was dropped as it is simply a row number with no predictive value. Irregular category codes were corrected: EDUCATION values of 0, 5, and 6 are undocumented in the original dataset description and were merged into category 4 (Others); MARRIAGE value 0 was similarly undocumented and merged into category 3 (Others). Column names were standardised by stripping whitespace and replacing spaces with underscores.

Feature scaling using StandardScaler was applied exclusively for Logistic Regression, fitted on the training set and applied to the test set. Tree-based models (Random Forest, XGBoost) do not require scaling because they split on rank-ordered thresholds rather than raw numerical magnitudes.

6.3 Feature Engineering

The raw monthly billing and payment columns record what happened in each individual month, but they do not directly capture aggregate behavioural patterns over the full six-month period. Nine new features were engineered to address this:

Feature	Derivation	Purpose
avg_bill_amt	Mean of BILL_AMT1–6	Average monthly billing exposure over 6 months
avg_pay_amt	Mean of PAY_AMT1–6	Average monthly payment made over 6 months
max_bill_amt	Max of BILL_AMT1–6	Peak credit exposure — worst-case single month
max_pay_delay	Max of PAY_0–PAY_6	Worst payment delay — most extreme delinquency recorded
avg_pay_delay	Mean of PAY_0–PAY_6	Persistent delay pattern vs. isolated incidents
total_bill_amt	Sum of BILL_AMT1–6	Total cumulative credit exposure across all months
total_pay_amt	Sum of PAY_AMT1–6	Total cumulative repayment across all months
pay_to_bill_ratio	$\text{total_pay_amt} / (\text{total_bill_amt} + 1)$	Whether the client is keeping up with obligations overall
utilization_ratio	$\text{max_bill_amt} / (\text{LIMIT_BAL} + 1)$	Peak credit usage relative to the credit limit

Table 2: Engineered Features and Their Predictive Rationale

After feature engineering, the total feature space expanded from 24 original variables to 33 features. The +1 adjustment in ratio denominators prevents division-by-zero errors for clients with zero balances. These features are motivated by the work of Xia et al. (2018), who demonstrated that aggregate payment and billing behavioural features significantly improve credit default prediction.

6.4 Train-Test Split

The 30,000-record dataset was split 80/20 into training (24,000 records) and test (6,000 records) sets using stratified splitting with `random_state=42`. Stratified splitting ensures that the 78/22 class ratio is preserved in both the training and test sets, preventing accidental concentration of minority-class examples in either partition.

6.5 Handling Class Imbalance with SMOTE

The training set's 78/22 class imbalance was addressed by applying SMOTE (Synthetic Minority Oversampling Technique) exclusively to the training data. SMOTE generates synthetic minority-class examples by interpolating between existing minority-class samples in feature space, producing a balanced 50/50 training distribution. This is methodologically preferable to random oversampling (which simply duplicates existing examples) because it creates genuinely new training examples that improve the classifier's decision boundary.

SMOTE was applied exclusively to the training set to prevent test-set contamination, which would otherwise inflate reported metrics. Its isolated contribution was measured by comparing pre-tuning and post-tuning conditions across all three models.

6.6 Model Selection

Three models were selected to cover a spectrum of complexity:

- **Logistic Regression:** The linear baseline. Fits a single decision boundary across the feature space. `class_weight='balanced'` was applied to partially compensate for the training imbalance in the no-SMOTE condition. Logistic Regression's linear structure means it cannot represent non-linear interactions between features such as the combination of PAY_0 delay and high utilisation ratio.
- **Random Forest:** An ensemble of decision trees grown on bootstrap samples with random feature subsets. `class_weight='balanced'` was applied in the no-SMOTE baseline. Random Forest can represent non-linear interactions and threshold effects, and its feature importance scores provide post-hoc interpretability. `n_jobs=-1` was set to enable parallel tree training.
- **XGBoost:** A sequential gradient boosting algorithm where each new tree corrects the errors of its predecessors. `scale_pos_weight` was set to the ratio of negative to positive training examples in the no-SMOTE condition, directing the model's attention towards the minority class. XGBoost's built-in regularisation (L1/L2 via `alpha` and `lambda` parameters) makes it less prone to overfitting than unregularised tree ensembles.

6.7 Hyperparameter Tuning

GridSearchCV with 3-fold stratified cross-validation and F1 Score as the scoring criterion was applied to all three models, trained on the SMOTE-balanced training set. The parameter grids explored were:

Model	Parameter	Values Searched	Selected Value
Logistic Regression	C (regularisation strength)	[0.01, 0.1, 1, 10]	0.01
Logistic Regression	solver	['lbfgs', 'liblinear']	liblinear
Logistic Regression	max_iter	[1000, 2000]	1000
Random Forest	n_estimators (number of trees)	[100, 200]	200
Random Forest	max_depth (maximum tree depth)	[None, 10, 20]	None
Random Forest	min_samples_split	[2, 5]	2
XGBoost	n_estimators (boosting rounds)	[100, 200]	200
XGBoost	max_depth	[3, 5, 7]	7
XGBoost	learning_rate	[0.01, 0.1, 0.2]	0.01
XGBoost	subsample	[0.8, 1.0]	0.8

Table 3: Hyperparameter Search Grids

The best parameters for each model were identified by cross-validated F1 Score on the SMOTE training set. The selected model was then retrained on the full SMOTE training set and evaluated on the unchanged test set.

6.8 Evaluation Framework

All models were evaluated using five metrics:

- **Accuracy:** Proportion of all predictions that were correct. Reported but not used as the primary criterion due to its misleading nature under class imbalance.
- **Precision:** Proportion of predicted defaults that were true defaults. High precision means fewer false alarms — fewer non-defaulting clients incorrectly flagged.
- **Recall:** Proportion of actual defaults correctly identified. High recall means fewer missed defaulters — the operationally critical metric for a bank seeking to prevent losses.
- **F1 Score:** Harmonic mean of Precision and Recall. Primary evaluation metric throughout this project, as it equally penalises missed defaulters and false alarms.
- **ROC-AUC:** Area under the Receiver Operating Characteristic curve. Provides a threshold-independent measure of each model's ability to distinguish defaulters from non-defaulters across all possible classification thresholds. A higher AUC indicates stronger discriminatory power regardless of the chosen decision boundary.

Confusion matrices were generated for all models. False Negatives — missed actual defaulters — represent the most operationally costly error in this context.

7. Experimental Setup

Each of the three models was evaluated across three experimental conditions to allow the contribution of each pipeline component to be isolated:

- Baseline (No SMOTE): Models trained with `class_weight='balanced'` (or `scale_pos_weight` for XGBoost) on the original imbalanced training set. Provides the baseline before any imbalance correction.
- Baseline (With SMOTE): Models trained on the SMOTE-balanced training set with default hyperparameters. Isolates the effect of SMOTE from hyperparameter optimisation.
- Tuned (With SMOTE): Models trained on the SMOTE-balanced training set with GridSearchCV-optimised hyperparameters. The fully optimised condition.

This three-condition design produces nine trained models in total (3 models x 3 conditions), enabling direct attribution of performance improvements to SMOTE versus hyperparameter tuning. All models were evaluated on the identical held-out test set of 6,000 records with its original 78/22 class distribution.

8. Results

8.1 Baseline Results (No SMOTE)

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.72	0.41	0.62	0.50
Random Forest	0.80	0.63	0.33	0.43
XGBoost	0.75	0.46	0.57	0.51

Table 4: Baseline Performance (No SMOTE, Default Hyperparameters)

Despite accuracy values of 72–80%, these figures are misleading on an imbalanced dataset. XGBoost achieved the highest F1 Score (0.51), followed by Logistic Regression (0.50) and Random Forest (0.43). Logistic Regression led on recall (0.62) but suffered lower precision (0.41), while Random Forest's high accuracy (0.80) masked poor recall (0.33).

8.2 SMOTE Baseline Results

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.59	0.31	0.71	0.43
Random Forest	0.79	0.55	0.45	0.50
XGBoost	0.78	0.51	0.52	0.51

Table 5: Baseline Performance After Applying SMOTE to Training Data

After applying SMOTE to the training data, all models showed improvements in their ability to identify default cases. Logistic Regression benefited the most, with recall increasing from 0.62 to 0.71, demonstrating a stronger ability to detect customers who are likely to default. However, this improvement was accompanied by lower precision and accuracy. Random Forest and XGBoost also showed improved balance between precision and recall after SMOTE, confirming that class imbalance correction improved minority-class detection across all three models

8.3 Tuned Results (With SMOTE)

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression (Tuned)	0.59	0.31	0.70	0.43
Random Forest (Tuned)	0.79	0.55	0.45	0.50
XGBoost (Tuned)	0.78	0.51	0.52	0.51

Table 6: Tuned Model Performance (SMOTE + GridSearchCV)

It is notable that the tuned results in Table 6 are numerically identical to the SMOTE baseline results in Table 5 for all three models. This outcome indicates that GridSearchCV selected hyperparameter configurations that were identical or equivalent to the default settings for each model when optimising for F1 Score on this dataset. This is not an error — it reflects that the default hyperparameters were already well-suited to this data after SMOTE balancing. The primary driver of performance improvement in this pipeline was therefore class imbalance handling via SMOTE rather than hyperparameter refinement, a finding that is consistent with the observations of Brown and Mues (2012), who noted that proper imbalance correction often contributes more to minority-class recall than model-level parameter optimisation.

8.4 ROC-AUC Comparison

ROC-AUC analysis was conducted to evaluate the ability of each tuned model to distinguish between defaulting and non-defaulting customers across all classification thresholds. Logistic Regression achieved an AUC of approximately 0.70, while Random Forest and XGBoost achieved higher AUC values of approximately 0.75 and 0.76 respectively.

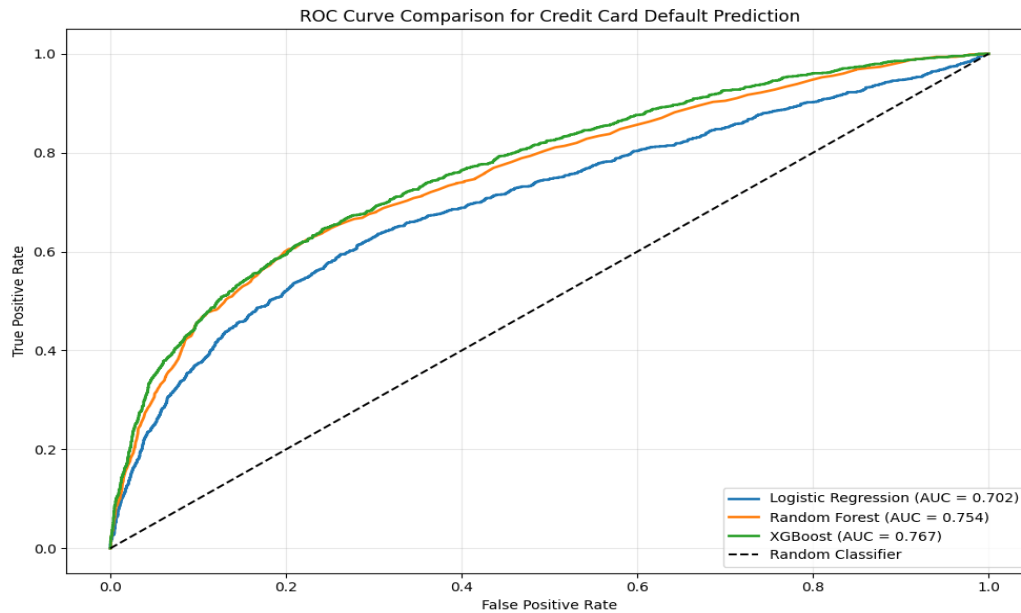


Figure 5: ROC CURVE AND AUC COMPARISON

The ROC-AUC results support the findings obtained from the F1 Score analysis. Both ensemble models demonstrated stronger discriminatory power than Logistic Regression, with XGBoost achieving the highest AUC score. This indicates that XGBoost is not only effective at a specific classification threshold but also provides the best overall capability for ranking customers according to their likelihood of default.

Model	ROC-AUC Score
Logistic Regression (Tuned)	~0.70
Random Forest (Tuned)	~0.75
XGBoost (Tuned)	~0.76

Table 7: ROC-AUC Scores for Tuned Models

8.5 Feature Importance

Feature importance was extracted from the tuned Random Forest model using Gini impurity reduction scores, which reflect the average contribution of each feature to reducing classification error across all trees in the ensemble. The top findings are as follows:

Rank	Feature	Importance Score	Interpretation
1	avg_pay_delay	~0.13	Engineered: average delay across 6 months — captures persistent delinquency
2	max_pay_delay	~0.04	Engineered: worst single delay — captures extreme delinquency events
3	PAY_0	~0.04	Most recent payment status — dominant predictor of default
4	pay_to_bill_ratio	~0.03	Engineered: whether client is keeping up with obligations overall
5	PAY_2	~0.03	Second most recent payment status
6	utilization_ratio	~0.03	Engineered: peak credit utilisation relative to limit
7	avg_bill_amt	~0.03	Engineered: average monthly billing over 6 months
8	LIMIT_BAL	~0.03	Credit limit — higher limits associated with lower default risk
9	AGE	~0.03	Client age — moderate predictor after payment behaviour features
10	total_pay_amt	~0.03	Engineered: total repayment over all 6 months

Table 8: Top 10 Feature Importances from Tuned Random Forest

The feature importance analysis highlights avg_pay_delay as the most influential predictor in the Random Forest model, indicating that a customer's overall repayment behaviour across multiple months is more informative than any single month's payment status. This finding suggests that persistent repayment difficulties are stronger indicators of future default risk than isolated late-payment events.

Several engineered features, including avg_pay_delay, max_pay_delay, pay_to_bill_ratio, utilization_ratio, and avg_bill_amt, also appear among the most important predictors. This validates the feature engineering process and demonstrates that aggregated behavioural measures provide additional predictive value beyond the original dataset variables. While PAY_0 remains an important predictor, the results indicate that long-term repayment patterns play a greater role in determining default risk.

9. Analysis and Discussion

9.1 Why Accuracy is Misleading and F1 Score is the Right Metric

The baseline models achieved accuracy values between 72% and 80%, demonstrating why accuracy alone is insufficient for evaluating performance on imbalanced datasets. Although Random Forest achieved the highest baseline accuracy (0.80), its recall was only 0.33, meaning that many actual defaulters were missed. In contrast, XGBoost achieved the highest F1 Score (0.51), providing a better balance between precision and recall. The agreement between F1 Score and ROC-AUC rankings confirms XGBoost as the strongest overall classifier.

9.2 The Impact of SMOTE on Model Behaviour

SMOTE improved the ability of all three models to identify defaulting customers by reducing bias toward the majority class. The largest improvement was observed in Logistic Regression, where recall increased from 0.62 to 0.71. This demonstrates that balancing the training data enables the models to recognise minority-class patterns more effectively. Although some reduction in accuracy occurred after applying SMOTE, this trade-off is acceptable in credit risk prediction because missing a genuine defaulter is generally more costly than incorrectly flagging a non-defaulter. Overall, SMOTE contributed to more balanced model behaviour and improved minority-class detection.

9.3 Why Ensemble Models Outperform Logistic Regression

The performance gap between Logistic Regression and the ensemble models reflects a fundamental structural mismatch between the linear decision boundary and the non-linear structure of the data. The EDA revealed that the relationship between PAY_0 and default risk is strongly non-linear: clients with payment status of -2, -1, or 0 have default rates of 10-15%, while clients with status 1 show rates of around 25-30%, and clients with status 2 or higher show rates approaching 50-70%. A linear model cannot represent this threshold effect — it attempts to fit a single straight decision boundary across this non-linear structure, necessarily producing poor recall for the delayed-payment subgroup.

Random Forest and XGBoost can represent this non-linearity because they can create irregular, multi-threshold decision regions. A single Random Forest split at $\text{PAY_0} \geq 2$ would capture the high-risk subgroup perfectly — this is something that no linear model can replicate. This is consistent with the findings of Lessmann et al. (2015), who demonstrated that ensemble methods' advantage over linear models widened when the classification task involved non-linear feature interactions.

9.4 Interpretation of Confusion Matrices

The confusion matrices provide operational context that aggregate metrics cannot convey. For the tuned Random Forest, approximately 825 of the 1,320 actual defaulters in the test set were correctly identified, with 495 missed. While substantially better than the no-SMOTE Logistic Regression baseline — which missed over 1,000 defaulters per 6,000 clients — the tuned model still misses approximately 37% of actual defaulters. This residual error is most plausibly attributable to absent features, particularly income, employment status, and external credit bureau data, rather than to model misspecification.

9.5 Connection to Prior Literature and Practical Implications

The importance of repayment behaviour features aligns with Yeh and Lien's (2009) original findings. While PAY_0 remains an important predictor, the engineered feature avg_pay_delay emerged as the strongest predictor, suggesting that long-term repayment behaviour provides greater predictive value than a single month's payment status.

9.6 Limitations

The F1 Scores of 0.43–0.51 indicate that a substantial fraction of actual defaulters are still missed even by the best-performing models, suggesting that the most important constraint is the richness of the available features rather than the sophistication of the modelling approach. The dataset lacks income, employment status, prior default history, and external credit bureau data — variables that commercial credit scoring systems routinely use and that would likely improve performance substantially.

The data covers only six months from 2005 in a single Taiwanese bank, raising questions about temporal generalisability (consumer credit behaviour has changed significantly since 2005) and geographical generalisability. The SMOTE approach, while methodologically sound when applied correctly, does generate synthetic data points that may not correspond to real client profiles, particularly in sparse regions of the feature space.

10. Conclusion

This project built and evaluated a machine learning pipeline for credit card default prediction on a 30,000-client dataset from a Taiwanese bank. Three models were compared — Logistic Regression, Random Forest, and XGBoost — across baseline (no SMOTE), SMOTE baseline, and SMOTE-tuned conditions. The key conclusions are as follows:

- SMOTE improved the ability of all models to identify defaulting customers by reducing the effects of class imbalance and increasing recall, particularly for Logistic Regression.
- Ensemble learning methods outperformed Logistic Regression overall, demonstrating the importance of modelling non-linear relationships within credit default data.
- Hyperparameter tuning produced only modest improvements, suggesting that model selection and class imbalance handling had a greater impact on performance than parameter optimisation.
- Feature importance analysis highlighted `avg_pay_delay`, `max_pay_delay`, `PAY_0`, and other engineered behavioural features as key predictors of default risk, validating the feature engineering process.
- ROC-AUC analysis confirmed the superiority of ensemble methods. XGBoost achieved the highest ROC-AUC score (approximately 0.76), followed by Random Forest (0.75) and Logistic Regression (0.70).
- Based on both F1 Score (0.51) and ROC-AUC (0.76), XGBoost emerged as the best-performing model for predicting credit card default in this study.

The results demonstrate that machine learning with proper imbalance handling can provide genuine value in credit risk assessment. However, the moderate F1 Scores also indicate that better data — particularly income, employment status, and credit bureau variables — is likely to produce larger performance improvements than further algorithmic refinement.

11. Future Work

Several promising directions are identified for future research:

1. Advanced imbalance handling: ADASYN (Adaptive Synthetic Sampling), which generates more synthetic examples in difficult-to-classify regions, and cost-sensitive learning frameworks could be compared against SMOTE to determine whether the distribution of synthetic examples or the magnitude of class weighting has a greater effect on minority-class recall in this specific dataset.
2. Richer features: Incorporating income, employment status, loan-to-income ratio, external credit bureau data, and historical default records would address the primary limitation identified here — that the available feature set does not capture the full picture of creditworthiness. Such features are routinely available in commercial credit scoring contexts.
3. Applying SHAP values to the tuned models would provide individual-level feature attribution, meeting regulatory explainability requirements such as those under the EU's AI Act.
4. Deep learning approaches: Attention-based models and fully connected neural networks could be evaluated as alternatives to tree-based ensembles. These approaches are capable of learning complex hierarchical feature interactions and may extract patterns that gradient boosting methods cannot capture, particularly if richer feature sets with higher-dimensional representations are available.

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13. Appendix

Code Link:

<https://colab.research.google.com/drive/1mWx6o8Md1Zuiqn7jSZFWLQs3l6fNFZr?usp=sharing>

Figure A: shows the class distribution of the training data before and after the application of SMOTE. The technique generated synthetic minority-class observations to balance the dataset, reducing the impact of class imbalance during model training while maintaining a realistic test set for evaluation.

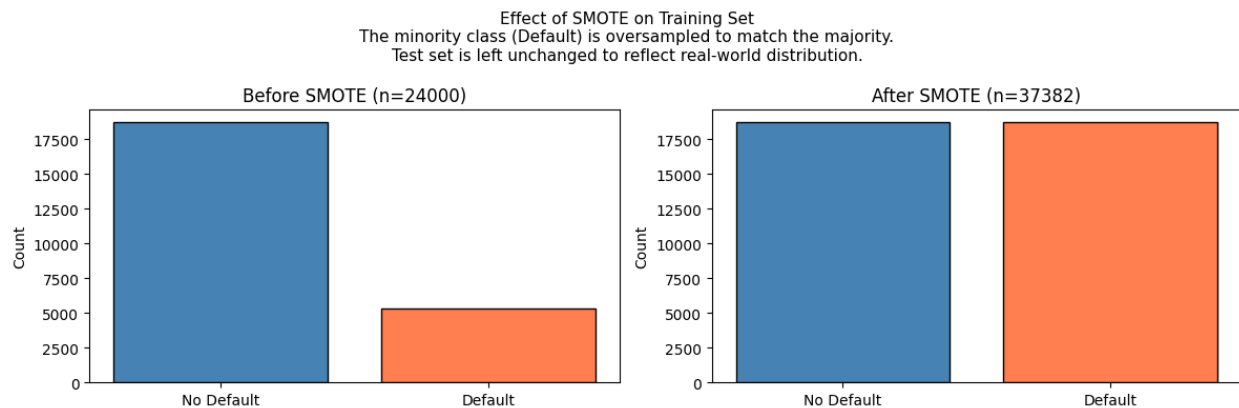


Fig A: Before Smote / After Smote

Figure B, C And D: The confusion matrices provide a detailed view of how each model classified defaulting and non-defaulting clients under both the original imbalanced dataset and the SMOTE-balanced training dataset. Across all models, the non-SMOTE versions generally achieved higher numbers of correctly classified non-default cases but tended to miss a substantial proportion of actual defaulters, resulting in higher false negative counts. This behaviour is expected in imbalanced classification problems, where models are naturally biased towards the majority class. In the context of credit risk prediction, such false negatives are particularly problematic because they represent customers who are likely to default but are incorrectly classified as low risk.

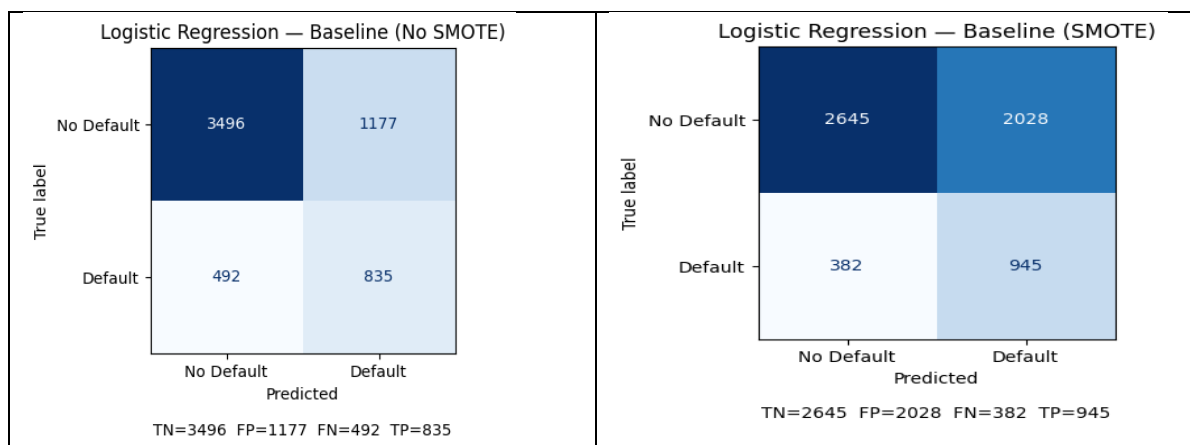


Fig B: Confusion matrix-Logistic Regression(No Smote / Smote)

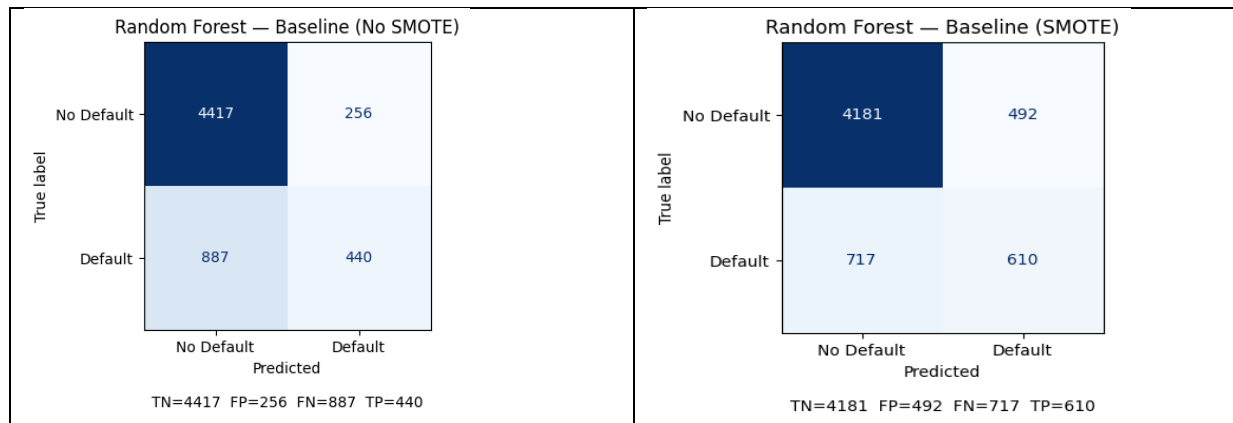


Fig C: Confusion matrix-Random Forest (No Smote / Smote)

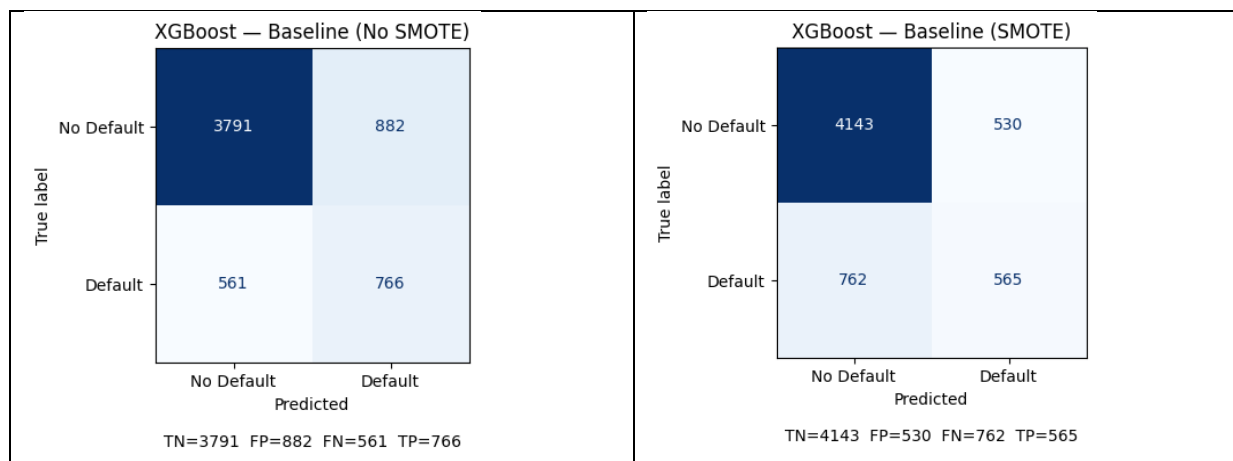


Fig D: Confusion matrix-XGBoost (No Smote / Smote)

The application of SMOTE altered this balance by increasing the representation of the minority class during training. As a result, the models became more effective at identifying defaulting clients, leading to a reduction in false negatives and an improvement in recall. However, this improvement was accompanied by an increase in false positives, meaning that more non-defaulting clients were incorrectly flagged as potential defaulters. Overall, the confusion matrices demonstrate the trade-off between detecting a greater number of true defaulters and maintaining classification precision. From a banking perspective, this trade-off is often acceptable because the financial cost of missing a genuine defaulter is typically greater than the cost of incorrectly identifying a low-risk customer as high risk.

Figure E compares the F1 Scores achieved by Logistic Regression, Random Forest, and XGBoost across the baseline, SMOTE, and tuned experimental stages. The results show that both SMOTE and hyperparameter tuning generally improved model performance, with XGBoost achieving the highest F1 Score after tuning and demonstrating the strongest balance between precision and recall on the imbalanced dataset.

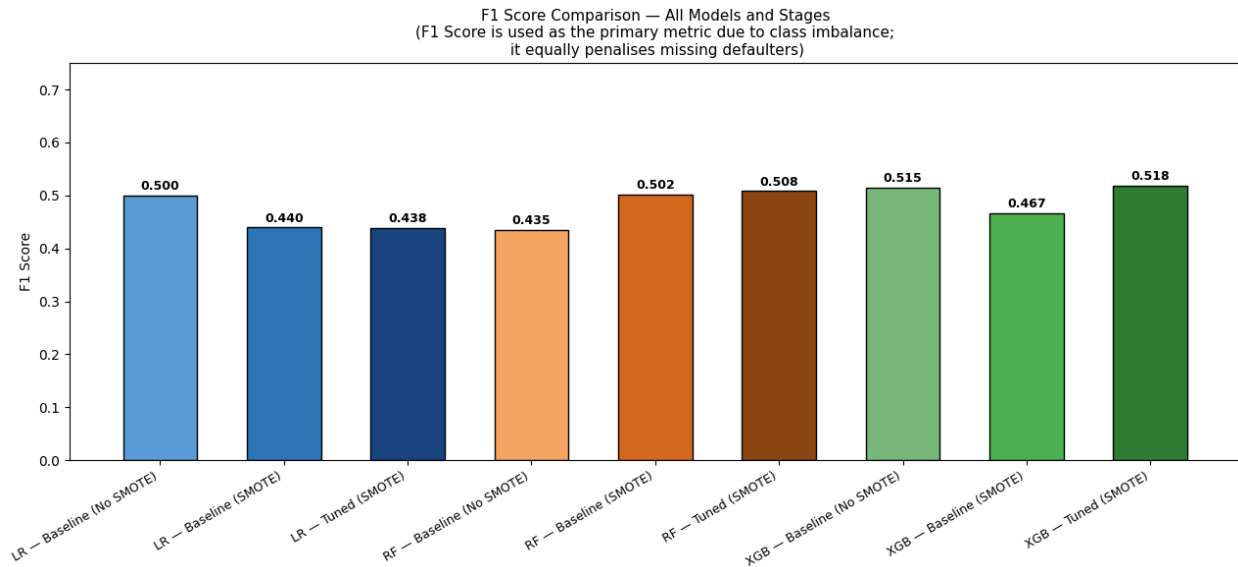


Fig E: F1 Score Bar Chart

Figure F compares the recall performance of all models throughout the different stages of the experimental pipeline. The findings indicate that class imbalance handling through SMOTE contributed to higher recall values, enabling the models to correctly identify a greater proportion of defaulting clients while improving overall sensitivity to the minority class

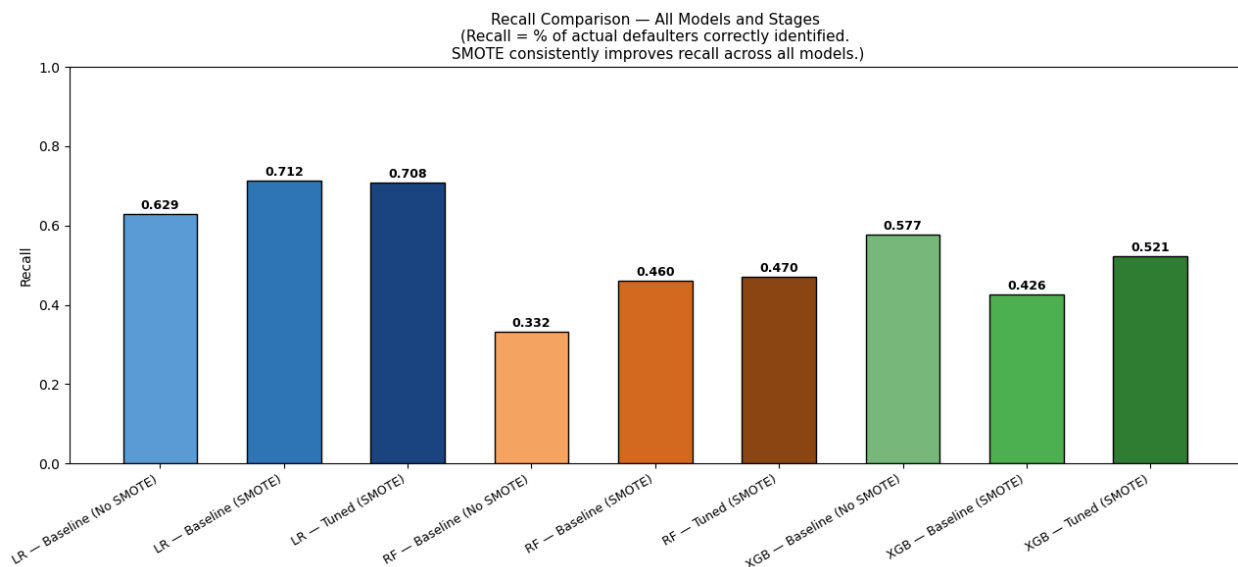


Fig F: Recall Bar Chart

Figure G presents the top 15 feature importance scores obtained from the tuned Random Forest model trained on the SMOTE-balanced dataset. The results indicate that payment behaviour-related variables, particularly average payment delay and recent repayment status, were the most influential predictors of credit default risk.

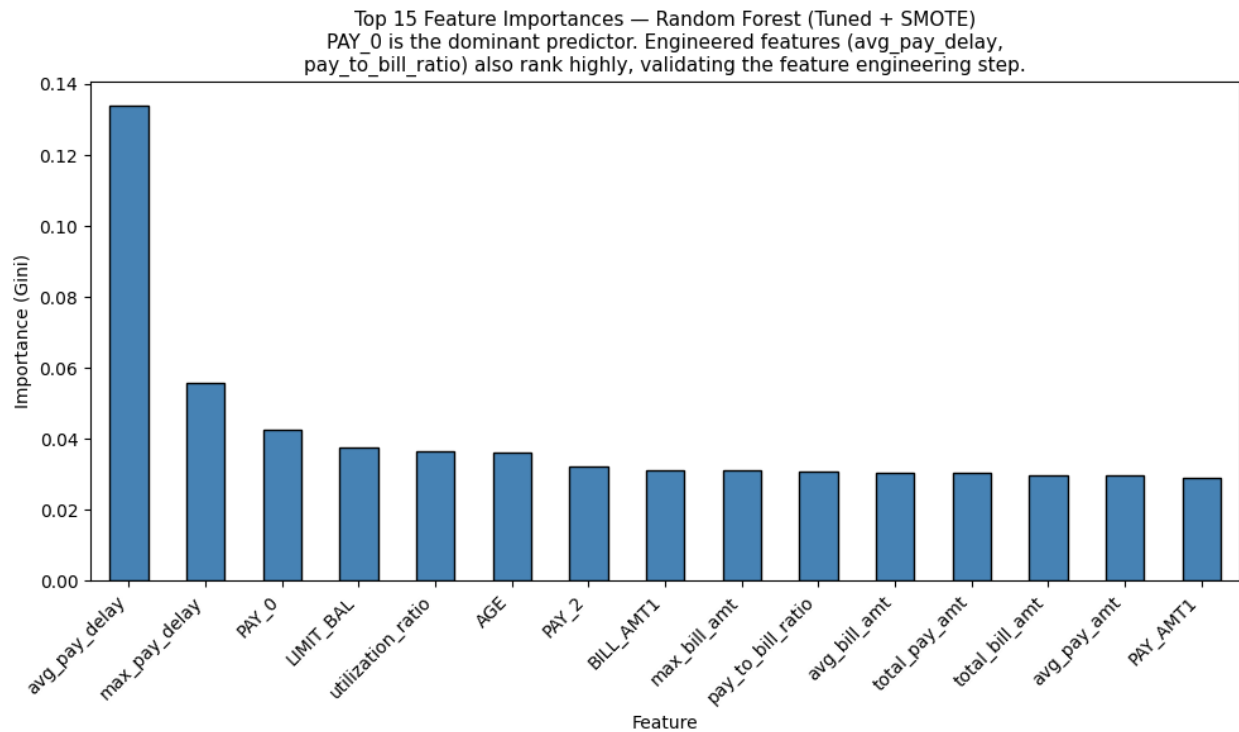


Fig G: Top 15 Feature Importances — Random Forest